

SL	Project Title	Project Idea
1	Analog to Digital to Analog: Signal reconstruction and visualization system	Convert a signal from Analog to digital, store it in a memory and then convert it again to an analog signal and finally display it on an oscilloscope.
2	Pulse width modulation with application	Students must design a pulse width modulator using op-amps with applications such as energy efficient lighting control based on sensors.
3	Schmitt Trigger based Noise Rejection Ckts or Level Detection based Control Ckts	Applications of Schmitt Trigger in handling noise. Design a schmitt trigger that will give correct output or level detection even if small noise are added to the input.
4	Digital Voltmeter (Without Arduino)	Students will design a voltmeter to read the analog voltage and display it digitally on a 7 segment display. Circuit must be built using ADC, DAC etc and without using arduino.
5	2 Bit ALU (Arithmetic Logic Unit) (Without Arduino)	ADD, SUB, MUL must, other operations optional. At least one operation must be carried out with logic circuits made using DL, RTL, DTL, TTL or NMOS instead of using ICs.
6	Digital thermometer	Students must design and implement a 4-bit Flash ADC based digital thermometer that converts temperature-dependent analog voltage into a binary digital output using only discrete analog and logic components (no microcontrollers). The output must be shown on a 7 segment display.
7	Electronic dice roller	Students must design and implement a dice roller that will display a random number from 0-6 on a seven segment display (no microcontrollers).
8	Electronic Piano	Students must design an electronic piano using 555 timer circuit.